

Limitations of Existing Solutions for Public Speaker Preparation

Project Context and desk research scope

This review answers the guiding question, **"What are the limitations of the solutions that already exist today?"**, within the context of a challenge statement is to build **an iOS app that helps public speakers become more prepared for public speaking or presentations.**

The scientific literature on this topic **unevenly distributed**. A large share of the evidence comes from higher-education settings, English-for-academic-purposes or second language presentation courses, and student populations rather than professional public speakers in workplace settings. A 2020 systematic review of VR for public speaking anxiety explicitly found that the most studied population was university students and concluded that the literature was still limited and inconsistent, while several more recent AI feedback studies also focused on undergraduates in course settings.¹

What the literature says current solutions already provide

The research does not say that current solutions are ineffective. In fact, several categories of solutions do help. VR-based interventions have shown meaningful anxiety reduction, and a large meta-analysis of 92 studies found VR to be a useful treatment approach for public-speaking anxiety, with efficacy comparable to other treatment methods in the studies reviewed. AI-supported tools have also shown positive effects in some educational contexts, for example, a 10-week mixed-methods study found significant improvement in public speaking anxiety reduction and oral-presentation skill when tools such as Yoodli and ChatGPT were used as scaffolded support. Peer assessment can also improve reflection and confidence, with classroom studies showing benefits for self-evaluation and presentation-skill development.²

So the core issue is that **today's solutions usually solve only part of the preparations problem**. Some tools reduce anxiety, some measure delivery mechanics, some help organize peer feedback, and some provide repeated practice opportunities. The evidence repeatedly shows that these solutions remain partial, narrow or context-bound rather than fully addressing what "being ready to present" actually requires.³

¹ <https://www.mdpi.com/2075-4426/10/1/14>

² <https://link.springer.com/content/pdf/10.1007/s12144-021-02684-6.pdf>

³ <https://staff.aist.go.jp/m.goto/PAPER/ICMI2007kurihara.pdf>

Where existing solutions still fall short

Fragmentation

Public-speaking readiness is multi dimensional, but many existing systems focus on just one slice of it. *Presentation Sensei* explicitly limited itself to “basic behavior patterns” because high-level semantics were difficult for computers to analyze, and it gave feedback on a small set of delivery indicators such as speaking rate, monotony, fillers, eye contact, and timing. The RAP system similarly focused on basic novice errors like posture, gaze, volume, filled pauses, and slide text. A 2025 Springer paper on AI speech feedback states this problem directly: existing speech-evaluation systems still tend to focus on isolated aspects rather than holistic feedback. OpenOPAF, although more comprehensive than many earlier systems, still centers on measurable multimodal analytics such as gaze, body language, volume, articulation speed, filled pauses, and slide text.⁴

Contextual and pedagogical feedback is still weak

Current tools are generally stronger at telling a speaker *what happened* than at explaining *why it matters* and *how to improve it in context*. In a 2023 study comparing AI and human scoring of oral presentations, AI fluency scores correlated weakly with human judgment because silence breaks alone missed rhythm, linking, stress, and the possibility that pauses may be purposeful rather than signs of poor fluency. The same study found that percentage-based emotion outputs were often less useful than qualitative human comments, because people notice the interaction of gestures, voice, energy, and audience awareness rather than facial expression alone. A 2025 study of EAP Talk similarly noted a practical limitation: the tool provided scores without detailed feedback and performed better on controlled speaking tasks than on uncontrolled presentation tasks.⁵

Difficulty of replacing social learning and human interpretation.

Practice with other people is valuable, but current solutions do not handle it well enough. SlideSpecs showed that presenters often receive feedback through scattered notes, emails, and conversations, and that organizing this feedback into actionable revisions is challenging. The same work found that verbal feedback has limited bandwidth because only one person can speak at a time, while public discussion can bias or inhibit other people’s comments. A 2024 quasi-experimental study comparing automated and peer assessment found that students in the peer-assessment condition reported significantly higher social engagement, suggesting that the interaction created by peer feedback cannot simply be replaced by automated scoring. A 2025 qualitative study also found that most students would still prefer a human public-speaking instructor to an AI coach.⁶

⁴ <https://staff.aist.go.jp/m.goto/PAPER/ICMI2007kurihara.pdf>

⁵ <https://www.mdpi.com/2071-1050/15/1/196>

⁶ <https://jeremywnr.com/papers/SlideSpecs-IUI-2023.pdf>

Anxiety support remains incomplete even when performance analytics improve.

VR solutions do reduce anxiety for many users, but the field still has evidence gaps. A 2023 Frontiers paper reviewing prior VR public-speaking studies noted that many earlier studies lacked proper control groups, often measured only one dimension of effectiveness, and rarely followed VR training with a real presentation in front of a real audience. The 2020 systematic review of VR treatment in students found only 13 studies with 481 students, described the literature as limited and inconsistent, and noted that most studies did not use random allocation. A 2024 Frontiers study on an open-access VR exposure platform reported improvement after a single 30-minute session, but the author also stated that the experiment was single-arm and that future work needed control groups, diverse demographics, and long-term follow-up.⁷

Many solutions are still prototype-heavy and evidence-light

The literature contains many feasibility studies, pilot studies, and small-sample evaluations, which are useful for innovation but weaker for broad conclusions. Examples in the papers reviewed here include eight practice talks in SlideSpecs, beta and reliability tests with 24 and 36 students in the MDPI AI platform study, a 29-participant single-arm VR experiment, and a 10-participant qualitative AI-coach trial. This does not invalidate the findings, but it does mean the field still lacks strong long-term evidence for sustained improvement, workplace transfer, multilingual performance, and cross-cultural generalization.⁸

The most defensible answer to the guiding question

Existing solutions are limited mainly because they are partial

Most current tools help with one or two subproblems of public speaking preparation, such as anxiety reduction, filler-word detection, pacing, slide critique, or peer reflection. The current solutions rarely combine the full set of needs a real speaker has: message structure, delivery quality, anxiety regulation, contextual explanation, audience adaptation, social rehearsal, and low-friction access in one coherent workflow.⁹

Many current solutions remain too shallow or too context dependent

AI systems often struggle with nuance and uncontrolled speaking. Peer and coach feedback are richer, but they are time consuming, socially biased and difficult to organize. VR helps with anxiety, but the evidence base still contains control-group, transfer, and accessibility gaps. The field has many useful tools, but no clearly dominant solution that is

⁷ <https://www.frontiersin.org/journals/virtual-reality/articles/10.3389/frvir.2023.1242544/full>

⁸ <https://jeremywnr.com/papers/SlideSpecs-IUI-2023.pdf>

⁹ <https://staff.aist.go.jp/m.goto/PAPER/ICMI2007kurihara.pdf>

simultaneously holistic, actionable socially aware, affordable, and strongly validated in real-life presentation contexts.

Conclusion

Based on desk research from scientific papers, the main limitation of existing solutions is that they usually address only part of the public-speaking preparation process. Some tools help reduce anxiety through VR, some provide automated feedback on measurable delivery features such as fillers, pace, gaze, or posture, and others support peer or teacher feedback. However, the literature shows that these solutions are still fragmented, often lack contextual and actionable feedback, struggle to evaluate authentic uncontrolled presentations, and do not fully replace the social value of human coaching or peer discussion. Research on VR and AI tools is promising, but much of it still relies on small samples, student populations, or limited experimental designs. Therefore, current solutions are useful, but they are not yet holistic, fully accessible, or strongly validated enough to solve the whole preparation problem for public speakers.